

INFINA

"I've always seen myself as something outside of this world. Something of a voyeur, I suppose. Not sexually, I don't mean, though I guess I've never really been brave enough to know." they thought.

A camera sat in one of Allo's claws which pointed out across Infina.

"How American... To name a planet after its economic potential."

The end one 9-inch long metallic finger punched down on the button and the camera let out a long wailing

"EEEEEEEEEEEE" as it charged up before the "KSH" sound of the shuttering flash.

Allo tightened the claw of their other, right hand, further ratcheting the fingers of the exoskeleton into the icy cliff.

"Allo!" A voice yelled from below. The voice itself was muffled and distorted, but a radio signal repeated and enhanced through Allo's suit. "I told you to warn me before you use that thing!!" The enhanced voice yelled. They were right to be angry, the flash of the camera was around six times brighter than a military flashbang.

Each photo sold for tens of thousands back home. This one would go for similar. A vortex of cloud in the upper left. A few bolts of Infina's rose lightning shot horizontally across the sky. Most of the light from the planet glowed up from cracks in the dark rock at the bottom of the ravine. Those were all unimportant features to photographing Infina though. The real money was in the Distortion. Its placement in the frame, its magnitude. When the first Rover sent back photos of Infina, the distortion was thought to be something ephemeral. A side effect of how the camera interacted with Infina's atmosphere. It didn't take too long to learn how mistaken that was.

"HEY!" the voice said over the intercom. "Can you even acknowledge what I said?" He was on Allo's side now. Five of his claws were sunk into the rock. Three cables from his chest shot out to latch in as well, and the toes of his feet drilled into the rock as well. With his free hand he flicked Allo's exoskeleton. The sound was partially masked by the sound of Allo's respirator pushing in fresh oxygen.

"Yeah." They responded. "My bad." And Allo continued to scale the cliff.

Infina wasn't habitable. It never would be. It was almost ironic how little things could last on a planet named "Infina".

Sure, maybe one day some eccentric quadrillionaire would plant a mansion down somewhere to appease their ego. Maybe it'd even be their employer, Elmer Collins. And he'd say it was "The first of many!". Like they're advancing the human frontier by living in an exotic villa. It was a delusion, but it can be frightening just how often delusions guide reality.

The pair continued scaling the rock. Built in Machine Learning helped guide their actions, identifying optimal movements, warning and correcting for inoptimal stabilities. All while providing heat, oxygen, and recycled water. Gravity on the planet was 1.2 times that of Earth's on average. According to Allo's mask they were keeping a 5 mph pace upward.

The HUD of Allo's mask identified the ice just above him with a dark green 99.8% climbing rating. Allo reached out to it and sunk his claws in, pulling himself up. Then to the

99.9%, then the 98.7%, the 99.2%, they continued climbing. Every so often the HUD would show a lighter green "90.2%" as the highest rated spot, or sometimes even lower. A yellow 63.4%. In these odd cases, Allo would pause just briefly to look about until the HUD identified a higher rated hold. It was a simple, methodical journey up the cliff.

Reaching the top, Allo vaulted over the lip of the cliff, the exoskeleton bracing a heavy landing onto a patch of dry ice, the pressure from his feet paused it's sublimation briefly, but shortly after landing it began puffing up once again.

"Warning." Allo said. Then came the "EEEEEEEEEEE, KSH!" as the camera charged up and shot a photo down the side of the cliff.

The photo captured the exoskeleton in motion. The way the claws dug into the rock. The shavings of ice falling off around it. The mask pushed a cloud of blue ozone, as it powered its way upward. Behind it was, of course, the dark craggy cliffside accented by the neon orange molten colors of cracks in the surface of the planet. Allo guessed it'd go for around 19,000\$.

"Bitch, what in the fuck are you doing?!" The exoskeleton said as it landed beside Allo.

"I tried to warn you. I needed the shot of the suit in motion."

"That shit's blinding! We're on a hostile planet for purpose, this isn't a vacation. We're not here to take cute pictures for social media." The man said, looking right at Allo. The HUD of Allo's mask identified him with a red 12% climbing rating.

"This is my purpose here." Allo replied. "You're the surveyor, I take the pictures."

"Yeah, but you've got to stop getting in the way of my work." He said, pushing Allo aside to keep walking. He headed in the direction of the Distortion.

"Have you ever thought your rules are getting in the way of my work? Ever considered what it takes to achieve a natural photo?" Allo said, but they'd turned off their intercom just before saying it. Then Allo followed behind the surveyor.

A surveyor. That's what he was. What they were. On long marches like this one, the exoskeleton's step-assist powered every motion. Allo didn't even have to think of what number to reach for, it was all somehow even more automatic than just walking and it gave Allo's mind the time to wander on its own.

Their destination was a smudge in the distance, but not a smudge by virtue of its distance. On the contrary, the Distortion was a vast thing. It contained all the same hues that Infina always had. The blacks, blues, browns, and pinks. But all those colors were blurred and bleeding, running together like some watercolor portrait.

Each day that smudge moved just a little bit, and back at the observatory they'd track its movement. Meteorologists shipped in cages just to watch a smudge. As the two marched, Allo continued to take his pictures.

A mile passed before the voice interrupted Allo's thoughts.

"I need you to stop with the photos for a moment while I get some readings." The man said over the intercom. "The stars are just starting to come out. My HUD's just barely able to detect them, but if we wait much longer there's no guarantee it'll get better than now." The man said, kneeling down and looking up at the distortion.

"Team Three, Day Sixteen, Log Six. This is Halliwa speaking, recording Celestial Reckoning of Distortion B6 as of seventeen twenty-four hours. Integrated almanac systems have picked up on the stars Alamus, Spica, Entaura, and Mimar. Automatic reduction tables have calculated a movement of 1.8 degrees south and 2.3 degrees west, albeit with only 64.3

percent confidence.” Halliwa reached around to a pack on the back of his exoskeleton to retrieve a device.

“EEEEEEEEEE, KSH” Allo took another photo. They’d save that one as “Surveyor”. Sometimes Allo felt their photography brought them just as close to the moment as actually being there. Like they could time travel through all these photos. Especially with a high definition XR-Reality mark-IV, like Allo had. There wasn’t much to the moment you couldn’t capture. In fact, there was more to the photo.

Maybe in the papers back home they’d only see what the eye could see. But when Allo got back to the observatory, loaded the photos up on his computer, he could zoom in. See Infina as it truly was, nearly down to the atomic level. Watch all the little atoms bumping around. The deterministic clockwork of it all. The camera could see even an iodine atom. Watch it pushed by a pressure gradient, find a slight magnetism with a piece of sodium, and snap together. Sometimes Allo wondered if they too saw percentages guiding their movements. “98.5, 97.6, 94.9...” as all the ions jostled around. Fighting for some pointless positioning.

The Surveyor couldn’t yell at Allo while he was recording. He was deterministic too. And if you took a picture of his brain you’d see it. One big iodine atom.

Halliwa placed a two foot long tripod down on the ground and pressed a button on the thing to begin its measurements.

“Gravimeter deployed.” Halliwa said to the recording. Meanwhile the gravimeter used a thousand lasers to track a thousand specks descending through density columns. It recorded the gravity to 3.678623512346812986712566972136986 g’s. The numbers went on and on well past the billionths of a percent, where the final digit of accuracy wobbled between 6, then 7, then 6, then 7, then 5, then 6, before a light turned green and the final digit settled on a 6. Halliwa’s exoskeleton HUD copied the number and attached it to the log, before standing and moving him to another coordinate to repeat the testing.

The exoskeleton made it obvious how deterministic they really were. Pushing him along, telling him where to go, You could feel that inevitability even without the exoskeleton if you really wanted. The impulses which guided your every action. The hunger before you eat. The anger before you lash out. Maybe you think you can resist those gradients, but can you really?

“Alright Allo, the gradient’s still weak here. It was a difference in the millionths. Or tens of millionths or something, I don’t know, it’s too many digits. Point is, let’s keep going.”

Allo watched as the exoskeleton packed away the gravimeter and kept marching. Allo’s exoskeleton followed.

As they walked the landscape began to change. Maybe not in ways the human eye could see, but Allo lived their life through a lens with thousands of times the magnification of the human eye. The average size of the rocks was smaller. The frozen flakes of ice and CO2 became increasingly packed together. The lightning grew brighter pink, the bolts shorter in length but more frequent.

“These were all deterministic too. Nothing unpredictable or unusual about it.” Allo thought, snapping a photo. The box let an extra long “EEEEEEEEEEEEEEEE” as as all the infinitesimal motors and subcameras of the advanced cameras found focus on every minute detail of the planet around them. Then came the shuttering “KSH” once it found focus.

“It’s amazing, isn’t it?” Halliwa’s voice came through the intercom. “I know you’re not much for talking, but we might really be able to do it today. We might make history. Be the first to

step foot into that thing. And hell, they might even be right about all its potential. We might really be doing something special today.”

“Maybe.” Allo said, turning towards the warbling blur ahead of them. Allo took another photo.

“Haha,” Halliwa chuckled at Allo’s response. “I just wish I didn’t have to work with so many damn eccentrics.” Shortly after saying it, he was blinded by Allo’s flash. A photo of him in his exoskeleton with the distortion and a blood orange sun peaking through clouds and half eaten by the blur in the background.

“I’m not the eccentric.” Allo thought to himself, watching Halliwa and his exoskeleton continue towards the distortion.

In some ways, Halliwa would probably agree with Allo about determinism. He was a scientist afterall. He’d recognize all the little rules and checklists that constructed them, but that’s as far as he’d take it. He only believed in determinism while he was in a laboratory or taking a physics exam. But when it came to himself? When they’re “his” achievements, “his” actions? Well, whenever he could say “I” he wasn’t so deterministic about it. There was pride.

There’s no skill in any dice roll, so why care if it’s your own? Even with this frontier. The exploration of Infina, of the Distortion. If they didn’t do it, another pair here would. If this team didn’t do it someone else would within a decade. There were millions of milestones along the way to Infina. Each achieved, in part, by thousands or tens of thousands or more people and supported by the whole rest of the billions of people on earth. It was all too arbitrary for pride.

The closer the pair of exoskeletons got to the Distortion the shorter their field of view grew. Blurred lines appeared around them. As if frozen trails of rain dotted the sky, translucent but diffracting the light. They shimmered in the air for a moment, before disappearing.

“EEEEEEEEEEEE, KSH” Allo captured a dense patch of them against a rough and grey backdrop. In the hours of marching, they had long since left the cinematics of the cliffside they’d ascended, but now, in the hazy distance, there seemed to be a second cliff to be ascended.

“What in the hell are these?” Halliwa said to himself, before he began to record. His camera looked out towards the distant dark orange sun which appeared smeared by thousands of translucent streaks of diffraction which separated it from them between them.

“It seems we have misjudged the nature of the Distortion.” Halliwa began the log. “What, at a distance, seemed to be a linear and uniform increase in gravity as we approached the Distortion, now appears hardly uniform at all.” Halliwa stated in awe of his surroundings. “We have followed the increasing gravity like a compass and it has led us to a storm.” Halliwa said, allowing the camera to take in the threads of gravity which speckled the atmosphere around him. Some drifted down, some stayed in place, other threads floated and a few poured.

“Are you seeing this Allo?!” Halliwa said “We’ve done it Allo! This is the Distortion!” Halliwa’s voice cheered over the intercom.

Allo only responded with the “EEEEEEEEEEEE, KSH” of another shuttering photograph.

“Allo, it is raining gravity! Don’t you see it? The way the atmosphere is being compressed and decompressed, it’s changing the refractive index!” Halliwa said, taking out the gravimeter and placing it on the ground.

As Halliwa set up the tripod-like device, Allo looked at their own exoskeleton’s hand and stuck out an 8-inch metal finger. A drop of gravity hit the tip of the finger, pushing the finger down before corrective measurements led the finger to bob up and down like a spring until it

stabilized. A haptic netting reflected the experience to Allo's real finger with a slight neural conduction.

"8.877534117888217320012066375245601 gs" the meter read. The number constantly fluctuated within a couple tenths of a g until a drop of gravity fell through the sensor, spiking the number up to 12. So-and-so or the occasional drop floated out of the ground and dropped the number down several gs from its resting average of around 8 gs. Halliwa recorded the gravimeter for some time.

Halliwa stood in meditation watching the numbers. He listened as his exoskeleton automatically inflated and deflated the internal g-suit to prevent blood from pooling. At this point, the suit was practically functioning as a second external heart, forming gentle waves, punching his blood through an electric peristalsis.

"It's time we head back." Halliwa heard over the intercom, breaking his meditation. "Our shift's set to end soon." Allo explained.

"Shift??" Halliwa replied. "Damn the shift, we're at the bleeding edge of everything! Not just of science, but of history! Of the Universe! We can't turn our backs on this!" Halliwa yelled over the intercom.

There was no sense in arguing with the intercom. Allo already knew how determined Halliwa was.

"Where to?" Allo asked.

"Up." The intercom replied.

"EEEEEEEEEE, KSH."

Allo captured the moment. An exoskeleton finger pointing out towards the cliff in a gravity rain.

Halliwa could picture the headlines. Like Washington Crossing the Delaware. "The Modern Pioneer!" *The Scientific Tomorrow* would read. Allo estimated it would sell for a million.

"The readings at the top will determine everything!" Halliwa cheered as his exoskeleton bounded towards the cliff and Allo followed.

Halliwa's exoskeleton hit the wall after a ten foot leap upwards. Six hooks ejected out from the chest towards the rock. Then it kicked the long spines of the boot into the rock before the claws followed suit. Halliwa said nothing, looking down to see Allo attached to the cliffside as well. Then a hook from the chest retracted back after being intercepted by a small drop of gravity and had failed to grip the wall. Once it had retracted it shot again, this time sinking in.

Halliwa wouldn't stop without getting readings from the top of the cliff. Allo didn't need to take a picture to see inside his head. There was a gradient he was following, like everything else. This time it was a cliff. But it was really a job. But really it was the glory. And it was all in his head.

Allo had earned some accolades in their time. They reached out for a 94.43% hold, partially digging into the rock. Sometimes there was money from achievement. Sometimes you get interviewed. Sometimes people cared. Most of the time they didn't.

Allo had, for some time, tried to live outside of those gradients, but you never could. No matter how hard you try, there's always something tugging at your mind. That's why Allo liked money. At least it was simple. The most tangible.

"EEEEEEEE, KSH" Allo shot a photo along the cliff, capturing the rain. It'd sell for 60 grand minimum.

"You know, Allo," the voice came over the intercom, "I always thought they called you the 'Dispassionate photographer' as a joke or something. It always seemed like an oxymoron to me. Dispassionate and Photographer. But here we are, in a place so impossible it might as well be magic. And yet you can't seem to see or feel that magic at all." the voice said.

"I just see things as they are." Allo replied.

"These things *are* magical, Allo!" it said.

Maybe that could be true, If you could forget all the suffering that brought us here.

"Do you ever get a dopamine hit from art?" Allo asked, following the numbers as they climbed. 92.31%, 86.42%,...

"Yes, all the time. Do you not?"

"I can for some art." Allo replied in their detached way. "But then, I get a bigger hit from porn... Isn't it funny? The first is 'beautiful' while the second's 'depraved'."

"You think a place like this?" The exoskeleton said, stopping to reach one claw out and gesture to the rain around them, "Is comparable to porn?"

"Yes." Allo said. "Especially for you." The voice nearly scoffed as its exoskeleton grinded against the rain to pull itself further up the cliff. "I've met a million people addicted to music. People who couldn't go a second in silence without aching for a distraction. People who can't even be left alone with their own mind."

"So I'm trapped on this planet with some sex junky? What is this?" The voice said.

"I don't give a fuck about porn." Allo said. "Just don't be such a hypocrite. You take so much pride in some addictions, some dopamine, some pleasures, and yet you're so disdainful of others."

The intercom fuzzed in response. Underneath it was the mechanical whirring of their exoskeletons crunching up the rock. 87.27%, 77.92%, 80.02%,...

"You know what the worst addiction is?" Allo said to the silence. "The most Human addiction? It's not porn, or music, or art, or nicotine. And yet it's all of them. It's *identity*. The addiction to seeing yourself as something more than a random splotch of chance. Everything evil stems from that. Every act of false justice, every errant belief. People construct a whole delusional world in their minds just to fit into it. Shoot themselves across the galaxy cause they're addicted to pretending it'll make them remembered."

"What I'm doing here has a purpose." The voice shot back. "Don't you forget, you're here too afterall. Taking little pictures to sell to the highest bidder and then talking down to me. It's pathetic."

"The money goes to charity." Allo said. Their exoskeleton bounded past Halliwa's. "Besides, I'm not the one that's upset. You're the one that devoted years of their life to being here. All for the glory of it. Not the art, not the beauty, not the "magic". If this place were really so magical to you it wouldn't matter if I liked it. But it's not. And now you're upset that I'm not cheering and approving and gratifying your ego like you wanted." Allo reached for the next hold to dig into.

74.67%... 67.52%....

Allo stopped. The numbers were yellow, so they looked around for a better hold.

64.98%, 52.37%...

The longer Allo looked, the more they found the same. But these holds should be as good as any of the others...

Allo felt the g-suit pressing the blood through their body.

"Let's turn back." Allo said through the intercom.

"Listen, thanks for your nihilistic little monologue, but this isn't just about me. If the readings at the top of this cliff come back right, it could truly change the world. And who gives a fuck if I take a little pride in that too? If we could just find a gradient strong enough to justify this mission, nothing else will matter." the voice returned. For the first time, Allo paid attention to the creaking sounds under the intercom's audio.

The numbers weren't falling due to the quality of hold or the type of rock, Allo realized. They were falling too consistently as they climbed. So... they had to be falling due to the gravity. Allo sunk one claw into the rock and shot out six hooks. Only two hooks made it to the cliff on the first shot.

Looking out from the cliffside confirmed it, what started as a rain had turned to a storm.

"Halliwa, stop." Allo repeated over the intercom. Their exoskeleton audibly strained against the tide of the storm.

"Someday you'll care, Allo." Halliwa said, unaware of the inevitable. "Just wait and see."

His exoskeleton carried on, practically swimming up the cliff, making large overhead movements before stabbing the claws into the rock and ratcheting further up.

Then, the dice roll didn't go Halliwa's way. The crack of the steel wasn't that much louder than the crack of rock had been. The claws of the boots sheared off and fell past Allo, and Allo could see what would come next.

"Allo." A voice said over the intercom. "Allo!" It yelled, just a moment later, as the failure at the boot cascaded through the exoskeleton. Two lines of hooks snapping first under the gravity. Then the only other hook that managed to dig in broke too. The other boot hadn't dug in enough to support weight. And once all the weight was on the claws, Halliwa watched as one stray drop of gravity pushed through his forehead and snapped the claws which attached him to the cliff.

The exoskeleton was designed to protect against falls. Under this gravity, it accelerated four times faster and to a higher terminal velocity.

"EEEEEEEEEEEE, KSH"

Allo snapped a photo as the exoskeleton fell by. That one would be worth a lot.

Elmer Collins was lying in bed orbiting the moon in a trillion dollar estate when he heard the news about his employee.

"Astronaut Halliwa Guire takes Fatal Fall shortly after discovering 'Gravity Rains'" the tag line read. There was a photo of an exoskeleton tumbling down a cliff, pushing its way through a neon blue gas. Chunks of rock fell around it. The grappling hooks attached to the chest of the thing trailed behind it pointlessly, waving in the winds of the fall.

Collins made a "tsk" sound to himself as he looked up at the news story projected on his ceiling. It was amazing how the media heard before him. Seconds later a message popped up.

"Sir, I'm sorry to report this, but..."

Collins dismissed the message for the moment.

"Gravity rain..." he thought. It fit the distant images of the Distortion, its characteristics. Apparently, he'd lost two rockets to rain.

"Billions of dollars washed away to a little drizzle?" he smirked at thought. But what was a billion dollars when compared to infinity?

Collins tightened his bath robe as he sat up. A sharkskin patterned white robe with pearlescent embroidery. The tag read: "Made in Luna."

Elmer stopped at the doorway from his bedroom.

"There will be a shuttle back to the moon this afternoon." He said it to the girl in his bed. She seemed to still be adjusting to the artificial gravity caused by the Estate's spin. The CEO continued to his simulation room.

A live view of Infina sprung in holograms from the floor of the simulation room. A composite of satellite perspectives. At the center sat the giant swirling Distortion. In the far left corner was a blip for the Observatory.

Collin raised his hand to a button on the wall. The word "Infrastructure" was written next to the button. When pressed, a series of digital structures popped into existence in a ring around the Distortion. Each on the border of the storm.

Perpetual Motion Machines. It was a physicist's Philosopher stone. An unobtainable dream. Something crackpots spent hundreds sending designs for to patent offices just for them to be thrown out. Something for magicians to show off while hiding a little battery to power the turning. Infina changed that. Machines which were once impossible were now inevitable.

Each of the projected machines was a simple loop. Something a child would play with. A ball made of lead the size of an apartment complex would roll into the Distortion. The amplified gravity would drive the thing to a terminal velocity down the tracks. Hitting the bottom, half its energy would be collected in a spring and fed through a generator for electric power. Then the ball would be redirected out of the Distortion where the ball, with its remaining energy and under a lesser gravity, would have the strength to roll all the way back up to its beginning.

Elmer smiled thinking of it. The simple beauty of a steel ball trapped in a loop. To think, all of humanity's development. Toiling away at frontier after frontier of mathematics, science, and engineering. Millenniums spent ripping away at these veils of ignorance. All of it just to arrive at something as simple as a ball on a loop.

Some people were like that too, really. Stuck in their little loops. Their lives were on tracks. They clawed their way up to something only to be slammed back down to the ground. But at least they generated a little excess wealth on the way down.

Maybe he was like that too, Elmer thought. Stuck in a loop. He nearly burst out laughing.

"It's just that my "loop" is in orbit around the moon!" He said to himself, taking a step closer to the digital array of Perpetual motion machines. As he stepped, a holographic mountain shuddered and was extinguished under the heel of his foot.

Poor... Halliwa, was it? When you're stuck in a loop, sometimes you don't survive the fall. That's just the nature of things.

Such a shame it was photographed too. It painted the picture so wrong. Really, the company was undershooting projected deaths by 15% this year. But people, in their little loops, wouldn't care for that statistic. They'd latch on to a photo of a mangled exoskeleton and scream of the tragedy. To them, this death was his fault, but in a decade or two, when their power bill fell by half, there'd be no "thank you" nor apology. Just money. And when did he spend that money? Well, they blamed him for that too. That was just what their little loops told them to do.

Elmer pulled up a holographic keyboard beneath his fingertips.

Dear Apeiron Employees,

It is with tears in my eyes that I must announce that our beloved Surveyor Halliwa Guire passed away in a tragic fall. Some of you may have already heard this news, especially due to the widely publicized images surrounding his death. Out of respect for him and his family, and concern for your own mental healths, it is my wish that employees avoid viewing these images if at all possible. Furthermore, all employees are to take a week off work to recover from the horrific tragedy of this moment. Halliwa was a beacon of light on a dark and cold planet. He will be greatly missed.

Sincerely yours,
Elmer Collins, CEO

The CEO wandered out from the simulation, tallying the price of a week's vacation in his mind. The price of a potential lawsuit. It would all be within budget, but his mind stopped calculating when he left the simulation room. Sitting on a couch by the orbital Estate's docking bay was the girl from before.

Collins pulled up the holographic keyboard once again. In the upper left hand corner he found a little slider labeled "1 g". He slipped it down until it hit 0.

The main estate sat on the outer edge of an incomplete circle. It was technically called an Orbital Arc-Habitat. A large mass called the Core orbited the moon. Two metal hallways acted as spokes connecting to the partial circle where the full estate was located and the Core spun, generating artificial gravity for the estate. If the circle were too much smaller, the speed of the spin would need to increase to nauseating degrees in order to compensate. But, if Collins had populated the rest of the arc with estates, well, then he would have had to share.

When Elmer dropped the slider to zero, the Core ejected its weight. Gravity-gradient tethers shot out into space from the core and from the surrounding arc. When the weight of the thing was pushed outward the spinning decelerated, much like how a figure skater might extend their arms to slow their spin, the Orbital Arc-Habitat did the same. Of course, it could not eject the momentum entirely, but it could get close.

"Ah!" The girl on the couch exclaimed in shock. A wave of nausea, as if on a roller coaster, hit her stomach, and she began to float off the couch.

"Change of plans, little miss." Collins said with a smirk, entering the doorframe, his toes floated an inch from the ground. "Why don't you stay a week extra?" he said, pushing off the doorframe with his hands and floating towards the couch.

"Men are so predictable." Veronica thought, as the CEO flipped through the air. He was desperately trying to slip his arm back into his tacky, almost blindingly white bathrobe. The sash had slipped out of the loops around his waist and slowly snaked away. The worst part was he tried to play it cool. "Isn't that something?" He asked, as if he meant to flip himself over.

"Ugh! It's all so amazing here!" she replied, as he squirmed in the air chasing the floating sash. "Why don't you show me around...?"

"Oh, babe, I had some other things i—" He said, his fingertip reaching the sash and dragging it back.

"Since I'm gonna be staying so long..." She finished her sentence.

"Well, of course..." the man said, threading the sash back into his bathrobe. When he'd managed to spin, roughly, rightside up he pulled up a menu amongst his fingers and ticked the gravity up to something slightly higher.

"Let's get started," he said as he gently floated to the floor. "I've got so much to show you." He said it with a smirk.

The gravity felt something akin to swimming. She'd kicked off against the couch to follow Elmer. The gravity wasn't that much different from the moon's. The walls on every lunar building had holds and other objects to help push off without waiting to land again. Here, there was only the occasional pole slicing through the large room to reach out to and push off of. Veronica grabbed one of these polls. She looked up to Elmer struggling in front of her, pumping his arms to keep his balance. She let go of the pole.

"The trick is, darling, to wade your arms through the air." He said seriously. "It helps with balance and speed."

"Oh really!? That's so helpful!" Veronica replied, waiting so she didn't pass him.

"You'll get used to it by the end of the week, I'm sure." He said, looking back. She mimicked him, paddling her arms through the air. "I think you'll find I live a life that's easy to get used to" He said chuckling. "Who knows... You may not want to go back..."

"Haha" she laughed. It was a reaction that had made her a lot of money.

At times she wondered if he knew how the world saw him. All clumsy and vulgar. Maybe he didn't know, just surrounded himself with sycophants. People like her. Paid to tell him what he wanted to hear. Paid to "Haha" inauthentically.

They kept paddling through the living room and into a long weight room curving around the edge of the Estate. A double paned window stared out into the blackness of space. On occasion a thick weighted rope stood out from and extended into the nothing.

"Check this out..." He said as he found his way to the end of a curved rack of dumbbells. Each weight was captured to the rack by a mechanical grip which released when he took hold of the 100 lb dumbbell. He curled it with one arm, the weight still too large for him to perform the rep cleanly.

"In this gravity, even you could lift this." He said the weight helped hold him to the ground. "It's maybe fifteen lbs equivalent in this gravity. BUT!" He said as he pulled up a menu, "if I just hit this button..."

Through the window the weighted cords retracted, like snakes recalled to their dens. And as they did, the spinning of the station quickened and the centrifugal force pulled the weight to the floor with a thump.

"I like to set the gravity to alternate cyclically with twenty pound weights. Make 'em weigh a bit more when I'm at the bottom of a DEEP stretch, really putting tension on those muscle fibers, and then I let the gravity decrease to help me finish the curl." He explained, picking up a twenty and curling it.

"Wouldn't it be just as good to curl a twenty-five pound weight all the way through?" She asked.

"No, no!" He said, walking down the rack and gesturing to a mirror. "This tells me exactly how to train!" he said, and he curled the weight once again. Over the mirror image of himself, a projection of his musculature popped over his body in blue. His heartrate appeared in the corner, a calendar of previous weights and growth rates. "Not everyone can afford the luxury of a perfect workout routine." The mirror couldn't see the CEO's muscle from beneath the fat.

Sometimes Veronica wondered if their relationship was as authentic as he could get. Sure, he had to pay her to be near him, but for the money at least he could speak his mind. Other people would change or grow as a person to accommodate relationships, but Elmer's mirror didn't seem to give him a daily routine for that kind of growth.

"Care to see more?" the CEO said, breaking her concentration. He led her past the variety of other exercise machines, walking on the curved window of the outer edge of the estate until they reached the next room.

This room was double sealed, with a small chamber before the main room with three steps up before the second door. This room was a poolroom. When the gravity was low the water floated up in big bubbles to float between. You splash a wave and it'd flow up to the ceiling and break into a smattering of dew drops. Some bubbles would be caught by the walls, sucked up into a pours and spit out again reheated.

So the two journeyed through the estate. For each room the CEO showed off the estate's largess. Each room equipped for every gravity. The kitchen had induction stoves for when low gravity prevented smoke from clearing. A trampoline room. A smoking lounge, where vents carried off the smoke in long, winding rivers in zero G. An arcade, wine cellar, climbing wall, and green room.

"All the money in the world and you spend it on parlor tricks in space." Veronica thought. But he must see himself like Willy Wonka.

"And this is my test hangar!" the CEO said, bringing her to a final, vast room. A row of flyers filled the room, each with wings and sails and "windpaddles". In the low gravity they rowed through the air, pushing the atmosphere behind them. He rode on one toy through the air yelling to her "This'll be the bike of some future day!" as six wide paper paddles stroked backwards in all directions and inched the man forward through the enormous hangar.

He showed it all off as if she were the lucky child who won a golden ticket. As if she weren't his ex-employee. As if she weren't trying to pay her way back to Earth.

Rainbow rain scattered off the Observatory's ceiling as speckles of stardust broke against it. Allo sat at a supercomputer in the lobby staring at the last photo of Halliwa before his death. Then clicked to the next photo, the first photo of Halliwa after his death.

A crumpled heap was indented into the rock at the bottom of the cliff. In the uneven gravity, some limbs hit the ground harder than others and the indentation reflected that fact. Now that Allo was back at the observatory, they could zoom nearly endlessly into the microscopic depths of his photos once again.

At two times the scale, the gravity rain became clearer. The strips of distortion looked like little ponds tracing their way through the air, blurring the world behind them. At ten times the scale, an asteroid belt of fine particles kicked up by Halliwa's landing became visible, and Allo had to steer through them to continue zooming. Beyond a hundred times of magnification, Allo began to see the grain structure of the exoskeleton's nanosteel. Sometimes the stuff was called "Penrose steel" after the precise atomic pattern of the superalloy. But Allo could pan around that and see through the face mask, zooming into the dead eyes of Halliwa.

The XR-Reality mark-IV Camera was not so simple a camera as to merely take a photo. Nor did it merely take a million subphotos at varying magnifications. Nor did it merely have the software to reconstitute a full image of those million subphotos. It didn't just capture additional X-Ray and infrared photos either. No. What was unique about the Mark-IV, was that it spat magnetic pulses in an electric echolocation to form a partial MRI. And it did all that in the span of a

"EEEEEEEEEE, KSH"

Allo dragged the image into custom software which took full advantage of the supercomputer he worked on to clean and conducted fourier analysis on the meta data of the image. Slice by slice it returned a full holographic image of Halliwa. Everything behind the Penrose steel of the exoskeleton was blurred at best, but the brain, at least the parts of it hidden behind the mask, were largely reconstructed in hologram.

The pattern that was Halliwa spread out before Allo. They turned it in the air, examining its nooks and crannies for the quality of hologram. As far as brain holograms went, the quality was poor, so they dragged in additional photos. Halliwa scaling their first cliff face, at the top of the cliff, photos from along the trek, and at the bottom of the next. Posing like a great explorer, Halliwa gave Allo a good shot of the side of his brain.

While the supercomputer chugged away at the data, Allo pulled up Apeiron's stock—

"Hey, Allo,..."

Allo hotkeyed away the tab upon hearing the voice. It was another employee, stepping out from behind the supercomputer's case. The computer was built bulkier than usual models, with additional casing and hard drive redundancies to accommodate the hundred-fold increased cosmic rays of Infina.

"We haven't really talked much, my name's Ell, but I just wanted to extend my condolences about Halliwa. I know you spent a lot of time with him, and it must be really difficult for you, especially being as separated from your friends and family as we are, so I and some of the others in my meteorology department made this for you." She said, handing him a folded piece of paper.

"Th-thanks." Allo said, and the woman hesitated a moment waiting for them to say a bit more, but, after a moment, left him alone to the humming of the supercomputer. Allo unfurled the paper.

Dear Allo,

We at the department of meteorology are wishing you well during this tough time. This planet can be a lonely place and we'd love to offer you a place to go if you should ever need someone to talk to.

With Love,

At the end of the page were the signatures and small personal notes of eight people, starting with "Ell" in yellow ink.

Allo stared at the paper for several seconds before a huff of liquid nitrogen through the supercomputer jolted them back to the moment. Allo checked over their shoulders before pulling up "Apeiron" once more. It was just as they expected.

On the day Halliwa's death was revealed to the public the media mourned the man. Stocks tumbled for the day by a thousand dollars. Credible reporting on the low feasibility of the Apeiron's promises. Now Apeiron's stock was in a rally, buying the dip after Halliwa's death. Buying the news that there was "gravity rain" as if that really changed anything about Apeiron's alleged business model. Two days after the public facing star of Apeiron plummeted to his death the CEO's estimated net worth rose by three hundred billion dollars.

To Allo, it was just math. Hardly even evil. You could watch the pieces fall into place, bounce off of one another, predict the pattern like a grand Newton's cradle. Watching it all mesmerized Allo, but the system was too big for anyone to stop. You jam yourself into the pattern, but the momentum would crush you.

Sometimes Allo imagined coexisting with the inevitable. Find an identity to get addicted to. Strive just for that. Start building your own career, do interplanetary interviews, be a face on every magazine. Accrue wealth. Donate a large enough fraction of your wealth for wikipedia to label you a "philanthropist". Get a hot wife, two kids, and a mansion on the moon. Do everything you can just to carve out an image of yourself that you can live with. Be like Halliwa. Halliwa would have bought the dip of his own death.

Or maybe Allo's identity could be having "hope". "Do their part". If you squinted your eyes at the logic of it, you could see some hope somewhere and cling to it. Then again, wasn't that just another identity? Another addiction? Another delusion where you could say "I'm making a real difference". Then you climb that ladder too, build up a little collection of social merit badges because you picked the more righteous delusion.

The problem is too many people root their identity in morality over truth. But what morality can there be without truth? Elmer Collins thinks his beliefs are "right and good". Thinks it's just the masses that hate him. Of course, his problem is just that he's a narcissistic moron who lacks the integrity to swallow any criticism or live in any reality that doesn't praise his efforts.

If you really cared about truth, they were all just identities. Stochastic atoms bouncing around, climbing up and falling down. Scattering into rainbows off observatory roofs. Better to have no identity. At least that was honest.

The whirring of the supercomputer softened, telling Allo its calculations were complete. The full recreation and diagnostics of Halliwa's brain. Only small details of memory could be resurrected from so few images. Small details were all Allo needed.

Samples of gravity were written into the pattern of nooks and crannies that were Halliwa. Just as Halliwa's brain read its own patterns to remember such details, so too could the

supercomputer read the hologram of Halliwa's brain and repeat the numbers back to Allo. There was loss, of course, but the most significant digits were preserved.

"8.8-7-3-----1----- gs" read the supercomputer, among a number of other readings collected by the gravimeter. These were all numbers Apeiron's higher-ups would have already known to some extent. They'd crashed hundreds of drones from orbit to learn the general figures. Elmer Collins even spoke at length in interviews about deducing the gravity of Infina by shooting down meteors with layered hulls. Each layer would be less and less heat resistant. Then you could derive the terminal velocity it'd reached, and therefore gravity, from how many layers burned away. For the design of the actual drones, they just used accelerometers. Funny they never did make the figures from those drones public.

The computer organized Halliwa's memories into folders for Allo to peruse. Schema for the perpetual motion machines were known to all employees on Infina. Lesser known were projections for getting the energy back to Earth.

From these figures, Allo began the calculations.

As planned by Apeiron, a 50 meter diameter steel sphere would dropped to accelerate under 8.8 gs to terminal velocity. The steel ball could accelerate to a terminal velocity of 890 meters per second in 27.3 seconds. To achieve 99% of that terminal speed the ball would need to drop from at least 18 kilometers or 11.2 miles up. In the process of falling the sphere would generate 2.03×10^{14} Joules. About 0.048 megatons TNT. The energy of a large tactical nuke.

In truth, letting the steel ball fall to terminal velocity was the least efficient way of siphoning power from the gravity distortion. It took disproportionately long to accelerate that last 5%. So much so that the energy to return the ball up that 18 kilometers, even under a lesser 3 gs, would far cost more energy than you'd accrue in the fall. About twenty-five percent more to be exact.

Better to start from a smaller drop height. The smaller the height, while still reaching into the highest gravities of the Distortion, would give the most net energy gain. To the supercomputer's calculation, the sweet spot, in theory, was around a 10 kilometer drop. The ball would reach 845 m/s, accrue 3.9×10^{14} Joules in kinetic energy in the fall, and use just 1.51×10^{14} of those Joules returning back to its starting point under 3gs. That left you with an extra 58% of your energy as profit per cycle. 2.39×10^{14} J. That'd run a large city for weeks.

The only problem was the losses. Efficiently catching the energy of an atomic bomb, replacing the parts from lightyears away, the degradation of the steel accelerating and braking over and over, and, of course, shipping the energy back to Earth.

As far as Allo could tell, that was the scam. Or, at least, one of them. Apeiron promised "Perpetual Power", Green energy to satiate any need for energy you could ever envision. Elmer promised an "If you build it, they will come" outlook on energy. "As the price of energy falls, the uses for energy rise" he'd argue. "Coal brought us steamboats and trains. Jetfuel brought us to the moon, and it was only once we'd embraced the Nuclear powers of the Atom that we have managed to reach Infina. Imagine what can be done with the infinite energy we find there?"

When pressed, Collins had answers for the "shipping problem". "We store the energy in cargo starships. We'll have infinite energy to spend on a conveyor belt of batteries bringing the energy back to Earth." He'd said one time. Another time he said "We'll beam the energy through a string of lighthouses across the stars". These claims didn't make much sense to the scientifically literate. The public wasn't scientifically literate. Nor were his investors.

It was easy to call Elmer Collins evil. Or an idiot. But more accurately, he was an addict. Addicted to an identity. To an image of himself where he was a genius superhero. Where humanity was fighting a cosmic war for survival and he was leading the crusade.

That's why Allo hated identities. You could make your identity anything, you could make it as noble and moral as you wished, if your identity wasn't founded in truth eventually you'd end up lying to yourself to maintain that identity. Elmer Collins was a stupid man with a uniquely stupid identity. A smarter man would hold truer identities. Get their identities from experts, have an identity in human-rights. The smartest man wouldn't have any identity at all. They'd never subject themselves to the tribalism inherent to identity.

The supercomputer huffed and Allo knew there was nothing they could do about Apeiron and its quadrillion dollar hoax. They stared down at the letter in their hands. An invitation to join them... Allo ripped the letter in half.

"Are you gonna need one of *these*?" Veronica said in a teasing voice, shaking a little bottle to show Elmer Collins as he reclined in bed.

"I'd never need something like that with someone as striking as you, babe..."

"Oh, but I *want* you to." She told him, turning and twisting the cap off with a *pop*. "I need you to *last*." She said, turning her head to wink. She didn't take a pill from the bottle, because she already had one in hand.

When Veronica made her way back to him she made sure not to walk. Instead she crawled over the bed to him.

"Lower the gravity, won't you?" She whispered to him, crawling up on top of him. "Make things easier on me..." She nuzzled up against his ear as she said "You're the only thing I want hard." and stuck her fingers, along with the pill, into his mouth.

"Yes, ma'am." Elmer said, swallowing with a gulp. Then he pulled up the projection of a menu as Veronica leaned back and popped the top button of her blouse.

"You're gonna do every single thing I say." She told him as he reached towards the menu's slider. His fingers fell before the gravity did, crashing against the sheets.

"Good boy." She told the unconscious CEO and rebuttoned her blouse.

The projected menu moved easily with her touch. Most features worked with her hand, but the occasional file or folder asked to scan his fingerprint. Luckily, she had those. She'd lift his limp hand for a laser from the ceiling to authenticate.

It wasn't hard to find what she wanted. All it took was a single query to his automated email AI.

"Did I deliberately have women fired from my Lunar headquarters soon after they were onboarded?" She typed in the interface. The Language Model thought for three seconds before it replied.

Great question!

...

Uh-oh!

Information relevant to your question is in a locked email. Would you like to conduct an eye-authentication? If not, I can also help by answering other questions! Would you like to hear about the history of debt-trapping? 🤔👉

Veronica popped the old man a second pill before answering. "Let's conduct that eye-authentication!"

She was not especially delicate when it came to pulling the middle-aged man's eyelids open. The hostage eyes stared blankly into the grid of lights which evaluated his irises. Eventually the model responded.

Great! Good to know its you Mr. Collins! ❤️

Now to your question.

Did you deliberately have women fired from your Lunar headquarters soon after they were onboarded?

Sort of!

What's True?

According to a string of emails from last November between you and Mr. Davis Gall, head of Apeiron's Lunar operations, you suggested that it "Would be nice if there were a few more female-led industries here on the moon" in response to Mr. Gall's email interested in downsizing lunar operations.

What's Not True

You did not directly cause the profit margins which motivated Mr. Gall's interest in downsizing. Nor did you explicitly tell Mr. Gall who to fire!

Hope that information was helpful! If you're interested in any other emails, let me know!



I can even reenact some of them if you wish! 🤖👉

The language model had attached a chain of emails relevant to its answer which Veronica downloaded and attached to an email sent to herself. Once she had those files downloaded on her phone she deleted the emails in his sent box. When Veronica had just about finished, a deep snore from the CEO startled her.

"You son of a bitch." She whispered. It reminded her of something: The rage that motivated what she was doing. Just for good measure, she began sending herself as much of his random data and emails as she could. Company projections, internal chatlogs, business decisions. She even asked the language model to help her find all the most incriminating data to send to herself. She took quite more pleasure in ripping the man's eye open for a second and third time, then pocketed a third pill in the corner of his cheek to slowly dissolve and keep him incapacitated.

Oh, and one more thing. She made him approve of the wire transfer he'd owed her. Ten million dollars. She'd need a hunk of it to afford her way back to Earth, but if she transferred too much more the bank would automatically flag the transaction. She found he'd already filled out the transaction form.

"Thank you...For your services ;)" It read.

"My pleasure." she thought to herself, approving the transfer.

Once she had everything, all she had to do was leave. With access to his estate controls, she didn't have to wait for the typical shuttle he made his "guests" use. Instead, she unlocked the door to his private Lunar transfer.

She left the menu open to the homescreen and made a dash through the simulation room towards the lobby. Her feet kicked through the topologies of that distant planet Infina. She blurred and scattered the projections as she passed, then entered the great lobby.

The massive room was easier to cross in standard gravity and without that deluded man slowing her down. She ran over windows pointing out towards the glimmering stars. She ran until she found a hatch which faced the same direction.

The hatch opened to a Roman interior. A white and gold circular fuselage, lavishly textured as though a Corinthian column had been inverted. This was, without question, the private shuttle of Elmer Collins. But she paused.

What if the man awoke before she'd landed? Could he control the shuttle remotely? Well, that was what the nearly lethal quantity of sleeping pills was for.

Once she reached Earth, she'd have everything she'd need to sue ever living fuck out of him. And who knows what else was in the data she'd taken. She crouched to descend into the fuselage, her foot finding a ladder which lined the side of the Roman column.

But then... How much could she sue him for? The word choice of the language machine popped into her mind. "Sort of". How many lawyers would Apeiron have? How much money could she even be awarded? A hundred billion? A trillion? It wouldn't even be one one-thousandth of that fuck face's worth. It wouldn't even be enough to pluck this estate from orbit.

And what had she done? What had she been forced to do? Just to afford living on a cold, pale, place. She'd abandoned her home to work a job that abandoned her. Veronica stepped out of the private shuttle's fuselage. There were too many things outside of her control in life. This wasn't one of them. Fuck him.

Veronica slammed the private shuttle's hatch shut and strode along the great lobby's floor. The glimmering stars wouldn't mind waiting just one more day. That man was still sound asleep with his menu in his bedroom. Veronica slid the gravity down to .1 gs. Then she kicked off the back of the bed and made her way to the weight room.

In .1 gs, the 100 lb weights were reduced down to ten. And before she returned to the bedroom, she found the time to practice her form in the mirror. Veronica pushed the projected menu down, away from the man, and sat with it on the drawer across from his bed. It was such a long wait for him to awaken, but some things were worth the wait.

When the CEO awoke he put his hand to his head and moaned. Across the room from him, she was floating in the air.

"Wh-" the man barely even managed before a 100 lb weight was flung through the air.

"Morning snookums." Veronica replied with a smile, then cranked the gravity to three times that of Earth's. With the command from the console, the weighted gravity-gradient tethers of the Quadrillionaire's Orbital Arc-Habitat whirled inward. And as they did, like a ballerina pulling her hands into a pirouette, the Estate began to spin faster and faster, and the weight grew heavier and heavier.

"EEKSH"

A three hundred pound weight cracked and sunk into the chest of Elmer Collins, CEO. When the blood sprayed outwards, it found that under the gravity it could only reach a third its usual height. The only thing left saturated in crimson was the sharkskin patterned, pearlescent embroidered robe with the tag "Made in Luna".

As for Veronica, when it came to a trial for the murder of Apeiron Industries CEO Elmer Collins. Well, at least it was her decision.